

Tutorial N° 2:(numeral systems)

Exercise1 :

1- Convert these binary numbers to decimal.

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|--------------|--------------|------------------|--------------|
| (a) 10110 | (b) 01101011 | (c) 1111010111 | (d) 10010101 |
| (e) 11111111 | (f) 11011111 | (g) 100100001001 | (h) 01101111 |

2- Convert the following decimal values to binary.

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|----------|----------|---------|---------|----------|
| (a) 37 | (b) 1000 | (c) 205 | (d) 13 | (e) 77 |
| (f) 2133 | (g) 189 | (h) 390 | (i) 511 | (j) 3410 |

3- Convert the binary numbers in Problem 1-1 to octal and hex.

Exercise2: Convert the following binary numbers to their equivalent decimal values.

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|---------------|-------------------|---------------------------|
| (a) 1100_2 | (b) 1001.1001_2 | (c) 10011011001.10110_2 |
| (d) 10011_2 | (e) 1100.0101_2 | (f) 10011100100.10010_2 |

Exercise3:

1- Convert each hex number to its decimal equivalent.

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|---------|----------|----------|----------|----------|
| (a) 743 | (b) 2000 | (c) 7FF | (d) 36 | (e) 55 |
| (f) 165 | (g) 1204 | (h) 37FD | (i) ABCD | (j) 000F |

2- Convert each of the following decimal numbers to hex.

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|---------|----------|------------|------------|----------|
| (a) 59 | (b) 1024 | (c) 65,536 | (d) 372 | (e) 771 |
| (f) 255 | (g) 919 | (h) 2313 | (i) 25,619 | (j) 2048 |

3- Take each four-bit binary number in the order they are written and write the equivalent hex digit

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|----------|----------|----------|----------|
| (a) 1001 | (e) 1111 | (i) 1011 | (m) 0001 |
| (b) 1101 | (f) 0010 | (j) 1100 | (n) 0101 |

4- Convert the hex values in Problem 3-1 to binary.

5- List the hex numbers in sequence from 280 to 2A0.

6- How many hex digits are required to represent decimal numbers up to 1 million?

Exercise4: By using the direct method:

A- Convert the following binary numbers to their hexadecimal and octal equivalents

- (a) 10100100 (b) 10100111 (c) 10100,010 (d) 10111,0101
 (e) 11011111000 (f) 11000,01011 (g) 110011100111 (h) 100110110,101011

B- Convert the fractional decimal numbers to binary, octal and hexadecimal forms:

- a- $(0.625)_{10}$ b- $(0.450)_{10}$ c- $(2.500)_{10}$
 d- $(5.1625)_{10}$ e- $(7.1875)_{10}$ f- $(0.350)_{10}$

Exercise5: Perform the following operations in binary

- $(1101010)_2 + (1110011)_2$ • $(1010)_2 + (111)_2$ • $(1010.011)_2 + (111.11)_2$
- $(111011)_2 - (11)_2$ • $(100001)_2 - (11110)_2$ • $(1001.1110)_2 - (11.1001)_2$

Additional exercise to look up:

Exercise1: Perform the following conversions in the corresponding bases.

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|--|---|
| $(87)_{10} = (?)_2 = (?)_8 = (?)_{16}$ | $(100111001)_2 = (?)_{10} = (?)_8 = (?)_{16}$ |
| $(328)_{10} = (?)_2 = (?)_8 = (?)_{16}$ | $(1111101)_2 = (?)_{10} = (?)_8 = (?)_{16}$ |
| $(1900)_{10} = (?)_2 = (?)_8 = (?)_{16}$ | $(10010101101)_2 = (?)_{10} = (?)_8 = (?)_{16}$ |
| $(3E)_{16} = (?)_2 = (?)_{10} = (?)_8$ | $(34)_8 = (?)_2 = (?)_{10} = (?)_{16}$ |
| $(234)_{16} = (?)_2 = (?)_{10} = (?)_8$ | $(567)_8 = (?)_2 = (?)_{10} = (?)_{16}$ |
| $(AF3)_{16} = (?)_2 = (?)_{10} = (?)_8$ | $(1534)_8 = (?)_2 = (?)_{10} = (?)_{16}$ |

Exercise2: Perform the following operations in the binary system, knowing that the numbers are represented on a byte (8 bits). In the case of division, specify the remainder

- | | |
|----------------------------|------------------------------|
| a) $10101_2 + 1010_2 =$ | f) $110111_2 - 100111_2 =$ |
| b) $10111_2 + 1010_2 =$ | g) $100011_2 \times 110_2 =$ |
| c) $110111_2 + 100111_2 =$ | h) $110110_2 \times 101_2 =$ |
| d) $10101_2 - 1010_2 =$ | i) $100101_2 \div 101_2 =$ |
| e) $10111_2 - 1010_2 =$ | j) $11010111_2 \div 110_2 =$ |